

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT
on

OPERATING SYSTEMS

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by Ramya J B (1BM25CS500), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026. The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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Index Sheet

Sl. No.	Experiment Title	Page No.
1.	Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time. a) FCFS b) SJF (non-pre-emptive) c) SRTF (SJF Pre-emptive) d) Priority (non-pre-emptive) e) Priority (pre-emptive) f) Round Robin	0-20
2.	Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue	21-25
3.	Write a C program to simulate Real-Time CPU Scheduling algorithms: a) Rate- Monotonic b) Earliest-deadline First c) Proportional scheduling	26-34
4.	Write a C program to simulate producer-consumer problem using semaphores	35-38
5.a	Write a C program to simulate the concept of Dining Philosophers problem.	39-42
5.b	Write a C program to simulate Banker's algorithm for the purpose of deadlock avoidance.	43-52
6.	Write a C program to simulate the following contiguous memory allocation techniques. a) Worst-fit b) Best-fit c) First-fit	53-58
7.	Write a C program to simulate page replacement algorithms. (Any one) a) FIFO b) LRU c) Optimal	59-67

Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

LAB PROGRAM – 1:

Question:

Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time. a) FCFS

b) SJF (non-pre-emptive)

c) SRTF (SJF pre-emptive)

d) Priority (non-pre-emptive)

e) Priority (pre-emptive)

f) Round Robin

a) FCFS:

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i;
    printf("\n\tNAME : R Nandini\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");
    printf("Enter number of processes: ");    scanf("%d", &n);

    int at[n], bt[n], ct[n], tat[n], wt[n], rt[n];
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    printf("Enter Arrival Time and Burst Time:\n");
    for(i = 0; i < n; i++)
    {
        printf("P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
    }

    ct[0] = at[0] + bt[0];

    for(i = 1; i < n; i++)
    {
```

```

        if(ct[i-1] < at[i])
ct[i] = at[i] + bt[i];
else
        ct[i] = ct[i-1] + bt[i];
    }

    for(i = 0; i < n; i++)
    {
        tat[i] = ct[i] - at[i];    wt[i] =
tat[i] - bt[i];    rt[i] = wt[i]; //
FCFS: RT = WT

        avg_tat += tat[i];
avg_wt += wt[i];    avg_rt
+= rt[i];
    }

    avg_tat /= n;
avg_wt /= n;    avg_rt
/= n;

    printf("\nProcess\tAT\tBT\tTAT\tWT\tRT\n");
for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], tat[i], wt[i], rt[i]);
    }

    printf("\nAverage Turnaround Time = %.2f", avg_tat);
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Response Time = %.2f", avg_rt);
return 0;
}

```

OUTPUT:

```
"C:\Users\Admin\Desktop\1W" X + v
NAME : R NANDINI      USN : 1BM25CS497      SECTION : '0'
Enter number of processes: 4
Enter Arrival Time and Burst Time:
P1: 0 2
P2: 1 2
P3: 5 3
P4: 6 4

Process AT      BT      TAT      WT      RT
P1      0      2      2      0      0
P2      1      2      3      1      1
P3      5      3      3      0      0
P4      6      4      6      2      2

Average Turnaround Time = 3.50
Average Waiting Time = 0.75
Average Response Time = 0.75
Process returned 0 (0x0)  execution time : 12.201 s
Press any key to continue.
```

b) SJF (Non Preemptive)

CODE:

```
#include <stdio.h>

int main() {
    int n,i,j;    int
    pid[10],at[10],bt[10],ct[10],tat[10],wt[10];    int
    finished[10]={0};    int
    current_time=0,completed=0,min;    float
    avg_tat=0,avg_wt=0;

    printf("\n\tNAME : R NANDNI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");
    printf("Enter number of processes: ");    scanf("%d",&n);

    printf("Enter Arrival Time:\n");
    for(i=0;i<n;i++)
    {
        pid[i]=i+1;
    }
    scanf("%d",&at[i]);
    }

    printf("Enter Burst Time:\n");
    for(i=0;i<n;i++)
    scanf("%d",&bt[i]);

    //Sorting according to arrival time    for(i=0;i<n-
    1;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(at[i]>at[j])
```

```

{
int
temp;

temp=at[i];
at[i]=at[j];
at[j]=temp;

temp=bt[i];
bt[i]=bt[j];
bt[j]=temp;

        temp=pid[i];
pid[i]=pid[j];        pid[j]=temp;
    }
}
}

//Scheduling
while(completed<n)
{
min=-1;

    for(i=0;i<n;i++)
    {
        if(at[i]<=current_time && finished[i]==0)
        {
            if(min==-1 || bt[i]<bt[min])
                min=i;
        }
    }
}

```

```

        if(min==-1)
        {
            current_time++;
        }
    else
    {
        ct[min]=current_time+bt[min];
        tat[min]=ct[min]-at[min];
        wt[min]=tat[min]-bt[min];

        current_time=ct[min];

        finished[min]=1;
        completed++;

        avg_tat+=tat[min];
        avg_wt+=wt[min];
    }
}

printf("\nSJF Non-Preemptive Scheduling\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

for(i=0;i<n;i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
}

printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

```

```
return 0;
}
```

OUTPUT:

```
"C:\Users\Admin\Desktop\1W" X + v

NAME : R NANDINI      USN : 1BM25CS497      SECTION : '0'

Enter number of processes: 4
Enter Arrival Time:
0 8 3 5
Enter Burst Time:
7 3 4 6

SJF Non-Preemptive Scheduling
PID    AT    BT    CT    TAT    WT
P1      0     7     7     7     0
P3      3     4    11     8     4
P4      5     6    20    15     9
P2      8     3    14     6     3

Average Turnaround Time = 9.00
Average Waiting Time = 4.00

Process returned 0 (0x0)  execution time : 25.889 s
Press any key to continue.
```

c) SRTF (SJF Preemptive)

CODE:

```
#include <stdio.h>

int main() {    int
n,i,time=0,completed=0;    int
pid[10],at[10],bt[10],rt[10];    int
ct[10],tat[10],wt[10];    int
idx,min_rt;    float
avg_tat=0,avg_wt=0;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");

    printf("Enter number of processes: ");
    scanf("%d",&n);

    printf("Enter Arrival Time:\n");
    for(i=0;i<n;i++)
    {        pid[i]=i+1;
    scanf("%d",&at[i]);
    }

    printf("Enter Burst Time:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
    rt[i]=bt[i];
    }
}
```

```

while(completed<n)
{
    idx=-1;
min_rt=9999;

    for(i=0;i<n;i++)
    {
        if(at[i]<=time && rt[i]>0 && rt[i]<min_rt)
        {
min_rt=rt[i];
idx=i;
        }
    }

    if(idx== -1)
    {
time++;    }
    else    {
rt[idx]--;
time++;

        if(rt[idx]==0)
        {
            ct[idx]=time;
tattat[idx]=ct[idx]-at[idx];
wt[idx]=tattat[idx]-bt[idx];
avg_tattat+=tattat[idx];
avg_wt+=wt[idx];
completed++;
        }
    }
}

```

```

    printf("\nSJF Preemptive Scheduling (SRTF)\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
            pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }

    printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
    printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

    return 0;
}

```

OUTPUT:

```

D:\OS Lab Programs\LAB PRI
NAME : R NANDINI          USN : 1BM25CS497        SECTION : 'O'

Enter number of processes: 4
Enter Arrival Time:
0 8 3 5
Enter Burst Time:
7 3 2 6

SJF Preemptive Scheduling (SRTF)
PID    AT    BT    CT    TAT    WT
P1      0     7     9     9     2
P2      8     3    12     4     1
P3      3     2     5     2     0
P4      5     6    18    13     7

Average Turnaround Time = 7.00
Average Waiting Time = 2.50

Process returned 0 (0x0)    execution time : 55.467 s
Press any key to continue.
_

```

d)Priority (Non-Preemptive)

CODE:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n,i;    int
```

```
    pid[10],at[10],bt[10],pr[10];    int
```

```
    ct[10],tat[10],wt[10];    int
```

```
    finished[10]={0};    int
```

```
    current_time=0,completed=0,max;
```

```
    float avg_tat=0,avg_wt=0;
```

```
    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d",&n);
```

```
    printf("Enter Arrival Time:\n");
```

```
    for(i=0;i<n;i++)
```

```
    {        pid[i]=i+1;
```

```
    scanf("%d",&at[i]);
```

```
    }
```

```
    printf("Enter Burst Time:\n");
```

```
    for(i=0;i<n;i++)
```

```
    scanf("%d",&bt[i]);    printf("Enter
```

```
    Priority:\n");    for(i=0;i<n;i++)
```

```
    scanf("%d",&pr[i]);
```

```

while(completed<n)
{
max=-1;

for(i=0;i<n;i++)
{
if(at[i]<=current_time && finished[i]==0)
{
if(max==-1 || pr[i]>pr[max])
max=i;
}
}

if(max==-1)
current_time++;    else
{
ct[max]=current_time+bt[max];
tat[max]=ct[max]-at[max];
wt[max]=tat[max]-bt[max];

current_time=ct[max];

finished[max]=1;
completed++;
avg_tat+=tat[max];
avg_wt+=wt[max];
}
}

```

```

    printf("\nPriority Scheduling (Higher number = Higher priority)\n");
    printf("PID\tAT\tBT\tPR\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            pid[i],at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
    }

    printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
    printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

    return 0;
}

```

OUTPUT:

```

NAME : R NANDINI          USN : 1BM25CS497          SECTION : 'O'

Enter number of processes: 5
Enter Arrival Time:
0 2 3 4 6
Enter Burst Time:
3 2 5 4 1
Enter Priority:
5 3 2 4 1

Priority Scheduling (Higher number = Higher priority)
PID    AT    BT    PR    CT    TAT    WT
P1      0     3     5     3     3     0
P2      2     2     3     5     3     1
P3      3     5     2    14    11     6
P4      4     4     4     9     5     1
P5      6     1     1    15     9     8

Average Turnaround Time = 6.20
Average Waiting Time = 3.20

Process returned 0 (0x0)   execution time : 94.714 s
Press any key to continue.

```

e)Priority (Preemptive)

CODE:

```
#include <stdio.h>

int main() {
    int n,i;    int
    pid[10],at[10],bt[10],pr[10],rt[10];    int
    ct[10],tat[10],wt[10];    int
    completed=0,current_time=0,min;
    float avg_tat=0,avg_wt=0;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");

    printf("Enter number of processes: ");
    scanf("%d",&n);

    printf("Enter Arrival Time:\n");
    for(i=0;i<n;i++)
    {
        pid[i]=i+1;
        scanf("%d",&at[i]);
    }

    printf("Enter Burst Time:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
        rt[i]=bt[i];
    }
```

```

    printf("Enter Priority:\n");
for(i=0;i<n;i++)
scanf("%d",&pr[i]);

while(completed<n)
{
min=-1;

for(i=0;i<n;i++)
{
if(at[i]<=current_time && rt[i]>0)
{
if(min==-1 || pr[i] < pr[min])
min=i;
}
}

if(min==-1)
current_time++;
else {
rt[min]--;
current_time++;

if(rt[min]==0)
{
ct[min]=current_time;
tat[min]=ct[min]-at[min];
wt[min]=tat[min]-bt[min];

completed++;

```

```

        avg_tat+=tat[min];
avg_wt+=wt[min];
    }
}
}

printf("\nPriority Scheduling (Preemptive) Higher the Number, Higher the Priority\n");
printf("PID\tAT\tBT\tPR\tCT\tTAT\tWT\n");

for(i=0;i<n;i++)
printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);

printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

return 0;
}

```

OUTPUT:

```

NAME : R NANDINI          USN : 1BM25CS497          SECTION : 'O'
Enter number of processes: 7
Enter Arrival Time:
0 1 3 4 5 6 10
Enter Burst Time:
8 2 4 1 6 5 1
Enter Priority:
3 4 4 5 2 6 1

Priority Scheduling (Preemptive) Higher the Number, Higher the Priority
PID    AT    BT    PR    CT    TAT    WT
P1      0      8      3     15     15      7
P2      1      2      4     17     16     14
P3      3      4      4     21     18     14
P4      4      1      5     22     18     17
P5      5      6      2     12      7      1
P6      6      5      6     27     21     16
P7     10      1      1     11      1      0

Average Turnaround Time = 13.71
Average Waiting Time = 9.86

Process returned 0 (0x0)   execution time : 54.150 s
Press any key to continue.

```

f)Round Robin

CODE:

```
#include <stdio.h>

int main() {
    int n,i;    int
    pid[10],at[10],bt[10],rt[10];    int
    ct[10],tat[10],wt[10];    int
    tq,current_time=0,completed=0;
    float avg_tat=0,avg_wt=0;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");

    printf("Enter number of processes: ");
    scanf("%d",&n);

    printf("Enter Arrival Time:\n");
    for(i=0;i<n;i++)
    {
        pid[i]=i+1;
        scanf("%d",&at[i]);
    }

    printf("Enter Burst Time:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
        rt[i]=bt[i];
    }
}
```

```

    }

    printf("Enter Time Quantum: ");
    scanf("%d",&tq);

    int done;
    while(completed<n)
    {
    done=1;

        for(i=0;i<n;i++)
        {
            if(at[i]<=current_time && rt[i]>0)
            {
done=0;

                if(rt[i] >
tq)
                {
current_time += tq;
rt[i] -= tq;                }
            else
            {
current_time += rt[i];
rt[i]=0;
ct[i]=current_time;
tat[i]=ct[i]-at[i];
wt[i]=tat[i]-bt[i];

```

```

        completed++;

    avg_tat+=tat[i];
    avg_wt+=wt[i];
    }
    }
    }

    if(done==1)
    {
        current_time++;
    }

}

printf("\nRound Robin Scheduling\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

for(i=0;i<n;i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
}

printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

return 0;
}

```

OUTPUT:

```
"C:\Users\Admin\Desktop\1W X + v
NAME : R NANDINI      USN : 1BM25CS497      SECTION : '0'
Enter number of processes: 5
Enter Arrival Time:
0 1 2 3 4
Enter Burst Time:
5 3 1 2 3
Enter Time Quantum: 2

Round Robin Scheduling
PID    AT    BT    CT    TAT    WT
P1      0     5    14    14     9
P2      1     3    12    11     8
P3      2     1     5     3     2
P4      3     2     7     4     2
P5      4     3    13     9     6

Average Turnaround Time = 8.20
Average Waiting Time = 5.40

Process returned 0 (0x0)   execution time : 15.305 s
Press any key to continue.
```

LAB PROGRAM - 2

Question:

Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

CODE:

```
#include <stdio.h>

int main() {
    int n,i;    int
    at[10],bt[10],rt[10],q[10];    int
    ct[10],tat[10],wt[10];    int
    tq,current_time=0,completed=0;

    float avg_tat=0,avg_wt=0;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");

    printf("Enter number of processes: ");
    scanf("%d",&n);

    printf("Enter Arrival Time:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&at[i]);
    }
```

```

    printf("Enter Burst Time:\n");
for(i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
rt[i]=bt[i];
    }

    printf("Enter Queue Number :\n");
for(i=0;i<n;i++)
    {
        scanf("%d",&q[i]);
    }

    printf("Enter Time Quantum: ");
scanf("%d",&tq);

while(completed < n)
    {
        int executed
= 0;

// FIRST: QUEUE 1 (ROUND ROBIN)

        for(i=0;i<n;i++)
            {
                if(q[i]==1 && rt[i]>0 && at[i] <= current_time)
                    {
executed = 1;
if(rt[i] > tq)

```

```

        {
current_time += tq;
rt[i] -= tq;
        }

else

        {
current_time += rt[i];
rt[i] = 0;

        ct[i] = current_time;
completed++;
        }
    }
}

if(executed == 0)
{
    // SECOND: QUEUE 2 (FCFS)

    for(i=0;i<n;i++)
    {
        if(q[i]==2 && rt[i]>0 && at[i] <= current_time)
        {
current_time += rt[i];
rt[i] = 0;

        ct[i] = current_time;
completed++;

```

```

        executed =
1;        break;
    }
}
}

if(executed == 0)
current_time++;
}

// calculating average TAT and WT
for(i=0;i<n;i++)
{
    tat[i] = ct[i] -
at[i];    wt[i] = tat[i] -
bt[i];

    avg_tat += tat[i];
avg_wt += wt[i];
}

printf("\nProcess\tQ\tAT\tBT\tCT\tTAT\tWT\n");

for(i=0;i<n;i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
i+1,q[i],at[i],bt[i],ct[i],tat[i],wt[i]);
}

```

```

printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
printf("\nAverage Waiting Time = %.2f\n",avg_wt/n);

return 0;
}

```

OUTPUT:

```

NAME : R NANDINI          USN : 1BM25CS497          SECTION : '0'

Enter number of processes: 4
Enter Arrival Time:
0 0 0 10
Enter Burst Time:
4 3 8 5
Enter Queue Number :
1 1 2 1
Enter Time Quantum: 2

Process Q      AT      BT      CT      TAT      WT
P1      1      0      4      6      6      2
P2      1      0      3      7      7      4
P3      2      0      8     15     15      7
P4      1     10      5     20     10      5

Average Turnaround Time = 9.50
Average Waiting Time = 4.50

Process returned 0 (0x0)   execution time : 24.717 s
Press any key to continue.

```

LAB PROGRAM – 3

Question:

Write a C program to simulate Real-Time CPU Scheduling algorithms: a)
Rate- Monotonic
b) Earliest-deadline First
c) Proportional scheduling

a) Rate- Monotonic

CODE:

```
#include <stdio.h> #include
<math.h>
//TO CALCULATE GCD
int gcd(int a, int b)
{   return (b == 0) ? a : gcd(b, a %
b);
}
//TO CALCULATE LCM
int lcm(int a, int b)
{   return (a * b) / gcd(a,
b);
}

int main() {
    int n, i;    int bt[10],
pt[10], rt[10];    int hyper,
time = 0;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter Burst Times:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &bt[i]);

        rt[i] = 0; // initially no job released
    }
```

```

    printf("Enter Periods:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &pt[i]);
    }

// Calculate hyperperiod = LCM of all periods
hyper = pt[0];    for (i = 1; i < n; i++)
{
    hyper = lcm(hyper, pt[i]);
}

printf("LCM=%d\n\n", hyper);

printf("Rate Monotone Scheduling:\n");
printf("PID  Burst  Period\n");    for (i =
0; i < n; i++)
{
    printf("%d    %d    %d\n", i+1,
bt[i], pt[i]);
}

// Utilization check
double U = 0;    for (i
= 0; i < n; i++)
{
    U += (double)bt[i] / pt[i];
}
double bound = n * (pow(2.0, 1.0/n) - 1);
printf("\n%.6f <= %.6f => %s\n", U, bound, (U <= bound) ? "true" : "false");
printf("Scheduling occurs for %d ms\n\n", hyper);

// Timeline simulation (block-style)
int current_running = -1;    while
(time < hyper)
{
// Release jobs        for (i = 0;
i < n; i++) {        if (time %
pt[i] == 0) {        rt[i] =
bt[i];
        }
    }
}

// Pick highest priority (smallest period)
int min_period = 9999, index = -1;    for
(i = 0; i < n; i++) {        if (rt[i] > 0 &&

```

```

pt[i] < min_period) {          min_period
= pt[i];          index = i;
    }
}
// Print only when event changes
if (index != current_running)
{
    if (index != -1)
    {
        printf("%dms onwards: Process %d running\n", time, index+1);
    }
else
    {
        printf("%dms onwards: CPU is idle\n", time);
    }
    current_running = index;
}
// Execute one unit if a process is running
if (index != -1)
{
    rt[index]--;
}

time++;
}

return 0;
}

```

OUTPUT:

```

NAME : R NANDINI          USN : 1BM25CS497          SECTION : 'O'
Enter number of processes: 2
Enter Burst Times:
20 35
Enter Periods:
50 100
LCM=100

Rate Monotone Scheduling:
PID  Burst  Period
1    20     50
2    35     100

0.750000 <= 0.828427 => true
Scheduling occurs for 100 ms

0ms onwards: Process 1 running
20ms onwards: Process 2 running
50ms onwards: Process 1 running
70ms onwards: Process 2 running
75ms onwards: CPU is idle

Process returned 0 (0x0)   execution time : 14.917 s
Press any key to continue.

```

b) Earliest-deadline First

CODE:

```
#include <stdio.h>

int main() {
    int n, i;

    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION : 'O'\n\n");
    printf("Enter number of tasks: ");
    scanf("%d", &n);    int bt[n], dl[n], p[n],
    rem[n];    printf("Enter Burst times:\n");
    for (i = 0; i < n; i++) {        scanf("%d",
    &bt[i]);        rem[i] = 0; // initially no job
    released
    }

    printf("Enter Deadlines:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &dl[i]);
    }

    printf("Enter Periods:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &p[i]);
    }

    printf("\nPID\tBurst\tDeadline\tPeriod\n");    for (i =
    0; i < n; i++) {        printf("%d\t%d\t%d\t\t%d\n", i + 1,
    bt[i], dl[i], p[i]);
```

```

    }

    int total_time = p[0];
    for (i = 1; i < n; i++) {
        if (p[i] > total_time)
            total_time = p[i];
    }

    printf("\nScheduling occurs for %d ms\n\n", total_time);

    int prev_idle = 0;

    for (int time = 0; time < total_time; time++) {
        for (i = 0; i < n; i++) {
            if (time % p[i]
                == 0) {
                rem[i] = bt[i];
            }
        }

        // Select task with earliest deadline
        int selected = -1;
        int min_deadline = 9999;
        for (i = 0; i < n; i++) {
            if (rem[i] > 0) {
                if (dl[i] < min_deadline) {
                    min_deadline = dl[i];
                    selected = i;
                }
            }
        }
    }
}

```

```

// Print timeline      if (selected == -1) {          if
(!prev_idle) {        printf("%dms onwards: CPU is
idle.\n", time);      prev_idle = 1;
    }
    } else {          printf("%dms onwards: Task %d is running.\n", time,
selected + 1);        rem[selected]--;          prev_idle = 0;
    }
}

return 0;
}

```

OUTPUT:

```

C:\Users\Admin\Desktop\1W
PID      Burst  Deadline  Period
1         3       7         20
2         2       4          5
3         2       8         10

Scheduling occurs for 20 ms

0ms onwards: Task 2 is running.
1ms onwards: Task 2 is running.
2ms onwards: Task 1 is running.
3ms onwards: Task 1 is running.
4ms onwards: Task 1 is running.
5ms onwards: Task 2 is running.
6ms onwards: Task 2 is running.
7ms onwards: Task 3 is running.
8ms onwards: Task 3 is running.
9ms onwards: CPU is idle.
10ms onwards: Task 2 is running.
11ms onwards: Task 2 is running.
12ms onwards: Task 3 is running.
13ms onwards: Task 3 is running.
14ms onwards: CPU is idle.
15ms onwards: Task 2 is running.
16ms onwards: Task 2 is running.
17ms onwards: CPU is idle.

Process returned 0 (0x0)   execution time : 33.521 s
Press any key to continue.

```

c) Proportional scheduling

CODE:

```
// Proportional Share Scheduling (Lottery Scheduling)
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
// Define Process structure
```

```
typedef struct {    char  
name[5];    int tickets;  
} Process;
```

```
int main() {    int n,  
total_tickets = 0;    float  
total_T = 0.0;
```

```
    printf("Enter the number of Processes: ");  
    scanf("%d", &n);
```

```
    Process p[n];    srand(time(NULL)); // Seed random  
number generator
```

```
    // Input process details    for (int i  
= 0; i < n; i++) {  
printf("\nProcess %d:\n", i + 1);  
sprintf(p[i].name, "P%d", i + 1);  
printf("Tickets: ");    scanf("%d",  
&p[i].tickets);
```

```

        total_tickets += p[i].tickets;
total_T += p[i].tickets;
    }

    printf("\n--- Proportional Share Scheduling ---\n");

    // Input scheduling time period    int m;
    printf("Enter the Time Period for scheduling: ");
    scanf("%d", &m);

    // Lottery scheduling simulation    for (int i = 0; i
< m; i++) {        int winning_ticket = rand() %
total_tickets + 1;        int accumulated_tickets = 0;
int winner_index = -1;

        for (int j = 0; j < n; j++) {
            accumulated_tickets += p[j].tickets;        if
(winning_ticket <= accumulated_tickets) {
                winner_index = j;        break;
            }
        }

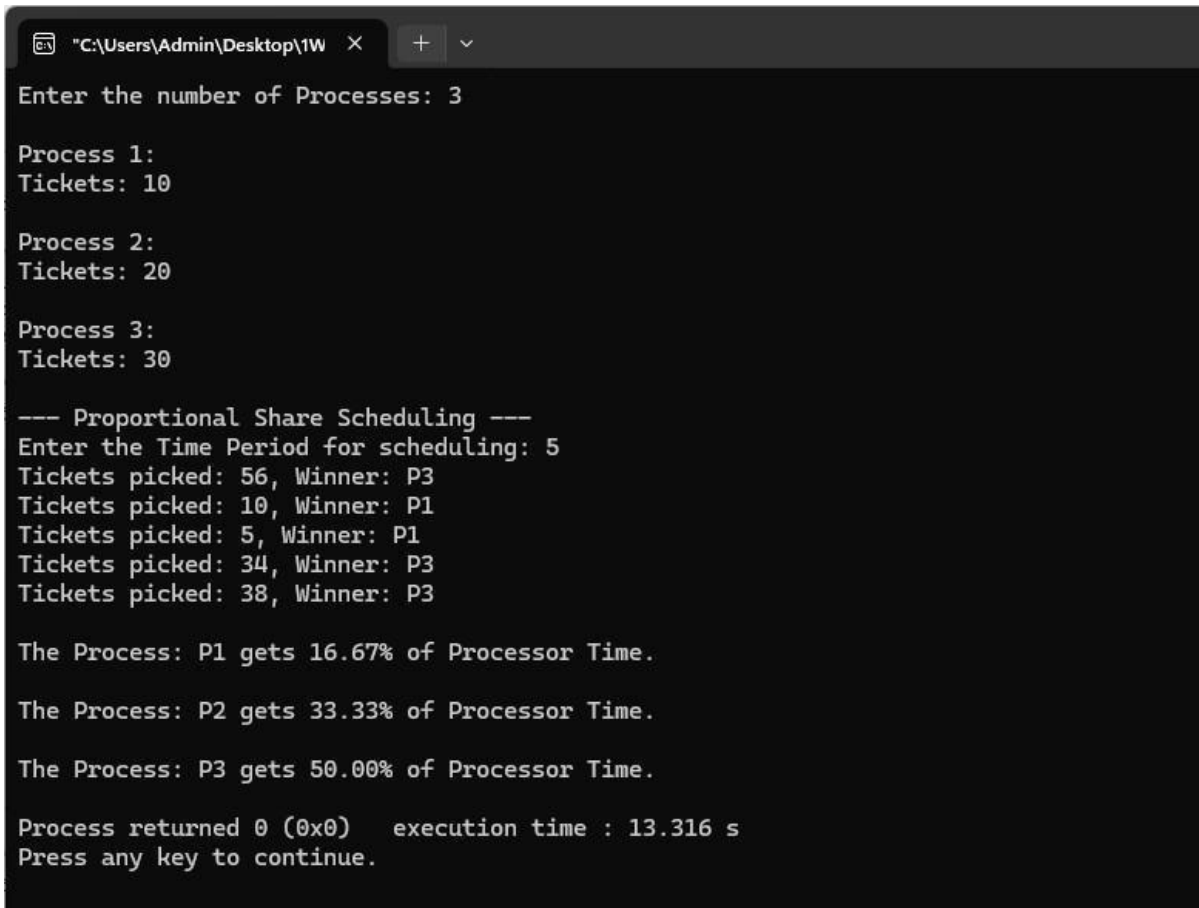
        printf("Tickets picked: %d, Winner: %s\n",
winning_ticket, p[winner_index].name);    }

    // Print proportional share of CPU time    for (int i = 0; i < n;
i++) {        printf("\nThe Process: %s gets %.2f%% of Processor
Time.\n",        p[i].name, ((p[i].tickets / total_T) * 100));    }

```

```
    return 0;
}
```

OUTPUT:



```
"C:\Users\Admin\Desktop\1W" X + v
Enter the number of Processes: 3

Process 1:
Tickets: 10

Process 2:
Tickets: 20

Process 3:
Tickets: 30

--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 56, Winner: P3
Tickets picked: 10, Winner: P1
Tickets picked: 5, Winner: P1
Tickets picked: 34, Winner: P3
Tickets picked: 38, Winner: P3

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.

Process returned 0 (0x0)   execution time : 13.316 s
Press any key to continue.
```

LAB PROGRAM – 4

Question:

Write a C program to simulate: (Any one)

- a) Producer-Consumer problem using semaphores.
- b) Dining-Philosopher's problem

a) Producer-Consumer problem using semaphores.

CODE:

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#include <semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0, out = 0;

sem_t empty; // counts empty slots sem_t full;
// counts filled slots pthread_mutex_t mutex; //
for mutual exclusion

// Producer function void*
producer(void* arg) {
    int item, i = 0; while (1) { item = i++; // produce an item
    (here just incrementing number) sem_wait(&empty);
    // wait for empty slot pthread_mutex_lock(&mutex); // enter
    critical section
```

```

        buffer[in] = item;    printf("Produced: %d at
buffer[%d]\n", item, in);    in = (in + 1) %
BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // leave critical section
sem_post(&full);              // signal that new item is available

        sleep(1); // simulate time to produce
    }
    return NULL;
}

// Consumer function void*
consumer(void* arg) {
    int item;  while (1) {    sem_wait(&full);          //
wait for filled slot    pthread_mutex_lock(&mutex); //
enter critical section

        item = buffer[out];    printf("Consumed: %d from
buffer[%d]\n", item, out);    out = (out + 1) %
BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // leave critical section
sem_post(&empty);          // signal that a slot is free

        sleep(2); // simulate time to consume
    }
    return NULL;
}

```

```

int main() {
pthread_t prod, cons;

    // Initialize semaphores and mutex
sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
pthread_mutex_init(&mutex, NULL);

    // Create producer and consumer threads
pthread_create(&prod, NULL, producer, NULL);
pthread_create(&cons, NULL, consumer, NULL);

    // Wait for threads to finish (they won't in this infinite loop)
pthread_join(prod, NULL);  pthread_join(cons, NULL);

    // Cleanup (not reached in this infinite loop)
sem_destroy(&empty);  sem_destroy(&full);
pthread_mutex_destroy(&mutex);

    return 0;
}

```

OUTPUT:

```
Produced: 0 at buffer[0]
Consumed: 0 from buffer[0]
Produced: 1 at buffer[1]
Produced: 2 at buffer[2]
Consumed: 1 from buffer[1]
Produced: 3 at buffer[3]
Consumed: 2 from buffer[2]
Produced: 4 at buffer[4]
Produced: 5 at buffer[0]
Consumed: 3 from buffer[3]
Produced: 6 at buffer[1]
Produced: 7 at buffer[2]
Consumed: 4 from buffer[4]
Produced: 8 at buffer[3]
Produced: 9 at buffer[4]
Consumed: 5 from buffer[0]
Produced: 10 at buffer[0]
Consumed: 6 from buffer[1]
Produced: 11 at buffer[1]
Consumed: 7 from buffer[2]
Produced: 12 at buffer[2]
Consumed: 8 from buffer[3]
Produced: 13 at buffer[3]
Consumed: 9 from buffer[4]
Produced: 14 at buffer[4]
Consumed: 10 from buffer[0]
Produced: 15 at buffer[0]
Consumed: 11 from buffer[1]
Produced: 16 at buffer[1]
Consumed: 12 from buffer[2]
Produced: 17 at buffer[2]
Consumed: 13 from buffer[3]
Produced: 18 at buffer[3]
Consumed: 14 from buffer[4]
Produced: 19 at buffer[4]
Consumed: 15 from buffer[0]
Produced: 20 at buffer[0]
Consumed: 16 from buffer[1]
Produced: 21 at buffer[1]
Consumed: 17 from buffer[2]
Produced: 22 at buffer[2]
Consumed: 18 from buffer[3]
Produced: 23 at buffer[3]
Consumed: 19 from buffer[4]
Produced: 24 at buffer[4]
Consumed: 20 from buffer[0]
Produced: 25 at buffer[0]
Consumed: 21 from buffer[1]
Produced: 26 at buffer[1]
Consumed: 22 from buffer[2]
Produced: 27 at buffer[2]
```

b) Dining-Philosopher's problem

CODE:

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

#define N 5 // Number of philosophers

pthread_mutex_t forks[N];    // One mutex per fork
pthread_t philosophers[N];   // Philosopher threads

// Philosopher routine void*
philosopher(void* num) {    int id =
*(int*)num;    int left = id;    // left fork
index    int right = (id + 1) % N; // right fork
index

    while (1) {    // Thinking
printf("Philosopher %d is thinking.\n", id);
sleep(1);

        // To avoid deadlock:
        // Even philosophers pick left then right    // Odd
philosophers pick right then left    if (id % 2 == 0) {
pthread_mutex_lock(&forks[left]);    printf("Philosopher %d
picked up left fork %d.\n", id, left);
pthread_mutex_lock(&forks[right]);    printf("Philosopher
%d picked up right fork %d.\n", id, right);
```

```

        } else {
            pthread_mutex_lock(&forks[right]);
printf("Philosopher %d picked up right fork %d.\n", id, right);

            pthread_mutex_lock(&forks[left]);
printf("Philosopher %d picked up left fork %d.\n", id, left);
        }

        // Eating
        printf("Philosopher %d is
eating.\n", id);
        sleep(2);

        // Put down forks
        pthread_mutex_unlock(&forks[left]);
pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put
down forks %d and %d.\n", id, left, right);
    }

    return NULL;
}

int main() {
    int i;
    int ids[N];

    // Initialize forks (mutexes)
    for (i = 0; i
< N; i++) {
        pthread_mutex_init(&forks[i], NULL);
        ids[i] = i;
    }

    // Create philosopher threads
    for (i = 0; i < N; i++) {
        pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
    }
}

```

```
}

// Join threads (infinite loop, but good practice)
for (i = 0; i < N; i++) {
    pthread_join(philosophers[i], NULL);
}

// Destroy mutexes (unreachable here)
for (i = 0; i < N; i++) {
    pthread_mutex_destroy(&forks[i]);
}

return 0;
}
```

OUTPUT:

```
Philosopher 0 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 4 is thinking.
Philosopher 3 picked up right fork 4.
Philosopher 1 picked up right fork 2.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 0 picked up left fork 0.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 1 put down forks 1 and 2.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 1 is thinking.
Philosopher 2 picked up left fork 2.
Philosopher 3 put down forks 3 and 4.
Philosopher 4 picked up left fork 4.
Philosopher 2 picked up right fork 3.
Philosopher 2 is eating.
Philosopher 3 is thinking.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
Philosopher 2 put down forks 2 and 3.
Philosopher 1 picked up right fork 2.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 2 is thinking.
Philosopher 3 picked up right fork 4.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 4 put down forks 4 and 0.
Philosopher 4 is thinking.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 2 picked up left fork 2.
Philosopher 1 put down forks 1 and 2.
Philosopher 1 is thinking.
Philosopher 3 put down forks 3 and 4.
Philosopher 3 is thinking.
Philosopher 4 picked up left fork 4.
Philosopher 2 picked up right fork 3.
Philosopher 2 is eating.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
```

LAB PROGRAM – 5

Question:

Write a C program to simulate:

- a) Bankers' algorithm for the purpose of deadlock avoidance.
- b) Deadlock Detection

a) Bankers' algorithm for the purpose of deadlock avoidance.

CODE:

```
#include <stdio.h>

int main()
{   printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION :
'O'\n\n");

    int n, m, i, j, k;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], max[n][m], need[n][m];
    int avail[m];

    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
    {   for(j = 0; j < m;
j++)
        {
            scanf("%d", &alloc[i][j]);
```

```

    }
}

printf("Enter Maximum Demand Matrix:\n");
for(i = 0; i < n; i++)
{
    for(j = 0; j < m;
j++)
    {
        scanf("%d", &max[i][j]);
    }
}

printf("Enter Available Resources:\n");
for(i = 0; i < m; i++)
{
    scanf("%d",
&avail[i]);
}

for(i = 0; i < n; i++)
{
    for(j = 0; j < m;
j++)
    {
        need[i][j] = max[i][j] -
alloc[i][j];
    }
}

int finish[n], safeSeq[n], work[m];

for(i = 0; i < n; i++)

```

```
    {    finish[i]
= 0;
    }
```

```
    for(i = 0; i < m; i++)
    {    work[i] =
avail[i];
    }
```

```
    int count = 0;
```

```
    while(count < n)
    {    int found
= 0;
```

```
        for(i = 0; i < n; i++)
        {    if(finish[i]
== 0)    {
int possible = 1;
```

```
            for(j = 0; j < m; j++)
            {
if(need[i][j] > work[j])
                {
possible = 0;
break;
                }
            }
        }
```

```
        if(possible)
```

```

        {
            for(k = 0;
k < m; k++)
        {
            work[k]
+= alloc[i][k];
        }

        safeSeq[count++] = i;
finish[i] = 1;        found =
1;
    }
}
}

if(found == 0)
{
break;
}
}

if(count == n)
{
    printf("\nSystem is in a safe
state.\n");    printf("Safe sequence is: ");

    for(i = 0; i < n; i++)
    {
        printf("P%d", safeSeq[i]);
        if(i != n -
1)    {
printf(" -> ");
        }
    }
}

```

```

    }
}
else
{
    printf("\nSystem is NOT in a safe state.");
}

return 0;
}

```

OUTPUT:

```

NAME : R NANDINI      USN : 1BM25CS497      SECTION : '0'

Enter number of processes: 5
Enter number of resources: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 52.958 s
Press any key to continue.

```

b) Deadlock Detection

CODE:

```
#include <stdio.h>

int main()
{   printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION :
'O'\n\n");
    int n, m, i, j, k;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    printf("Enter the number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], request[n][m];
    int avail[m];

    printf("Enter the allocation matrix:\n");

    for(i = 0; i < n; i++)
    {   printf("Process %d:
", i);

        for(j = 0; j < m; j++)
        {
            scanf("%d", &alloc[i][j]);
        }
    }
}
```

```

printf("Enter the request matrix:\n");

for(i = 0; i < n; i++)
{
    printf("Process %d:
", i);

    for(j = 0; j < m; j++)
    {
        scanf("%d", &request[i][j]);
    }
}

printf("Enter the available resources: ");

for(i = 0; i < m; i++)
{
    scanf("%d",
&avail[i]);
}

int finish[n], safeSeq[n], work[m];

for(i = 0; i < m; i++)
{
    work[i] =
avail[i];
}

for(i = 0; i < n; i++)
{
    finish[i]
= 0; }

```

```

int count = 0;

while(count < n)
{
    int found
= 0;

    for(i = 0; i < n; i++)
    {
        if(finish[i]
== 0)
        {
            int possible = 1;

            for(j = 0; j < m; j++)
            {
                if(request[i][j] > work[j])
                {
                    possible = 0;
                    break;
                }
            }

            if(possible)
            {
                for(k = 0;
k < m; k++)
                {
                    work[k]
+= alloc[i][k];
                }

                safeSeq[count++] = i;
                finish[i] = 1;
                found =
1;
            }

```

```

        }
    }

    if(found == 0)
    {
break;
    }
}

if(count == n)
{
    printf("\nSystem is in safe
state.\n");    printf("Safe Sequence is:
");

    for(i = 0; i < n; i++)
    {
        printf("P%d ", safeSeq[i]);
    }
}
else
{
    printf("\nDeadlock detected.\n");
}

return 0;
}

```

OUTPUT:

```

NAME : R NANDINI          USN : 1BM25CS497          SECTION : '0'

Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0

System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1

Process returned 0 (0x0)    execution time : 61.305 s
Press any key to continue.
```

LAB PROGRAM – 6

Question :

Write a C program to simulate the following contiguous memory allocation techniques. a)

Worst-fit

b) Best-fit

c) First-fit

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{    printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION :
```

```
'O'\n\n");
```

```
    int b[20], p[20];
```

```
int temp[20];    int
```

```
i, j, n, m;    int
```

```
allocation[20];
```

```
    printf("Enter number of memory blocks: ");
```

```
scanf("%d", &n);
```

```
    printf("Enter sizes of %d memory blocks:\n", n);
```

```
for(i = 0; i < n; i++)
```

```
{    scanf("%d",
```

```
&b[i]);
```

```
}
```

```
    printf("Enter number of processes: ");
```

```
scanf("%d", &m);
```

```

    printf("Enter sizes of %d processes:\n", m);
for(i = 0; i < m; i++)
    {
        scanf("%d",
&p[i]);
    }

// FIRST FIT

for(i = 0; i < n; i++)
temp[i] = b[i];

for(i = 0; i < m; i++)
allocation[i] = -1;

for(i = 0; i < m; i++)
    {
        for(j = 0; j < n;
j++)
            {
                if(temp[j]
>= p[i])
                    {
allocation[i] = j;
temp[j] = -1;
break;
                    }
            }
    }

printf("\n--- First Fit ---\n"); printf("Process
No.\tProcess Size\tBlock No.\n"); for(i = 0; i <
m; i++)

```

```

        {      printf("%d\t\t%d\t\t", i + 1,
p[i]);

        if(allocation[i]      !=      -1)
printf("%d\n", allocation[i] + 1); else
printf("Not Allocated\n");
    }

// BEST FIT

    for(i = 0; i < n; i++)
temp[i] = b[i];

    for(i = 0; i < m; i++)
allocation[i] = -1;

    for(i = 0; i < m; i++)
    {      int best
= -1;

        for(j = 0; j < n; j++)
        {      if(temp[j]
>= p[i])
            {
                if(best == -1 || temp[j] < temp[best])
                {
best = j;
                }
            }
        }
    }

```

```

        if(best != -1)    {
allocation[i] = best;
temp[best] = -1;
        }
    }
}

```

```

printf("\n--- Best Fit ---\n");   printf("Process
No.\tProcess Size\tBlock No.\n");

```

```

for(i = 0; i < m; i++)
{
    printf("%d\t\t%d\t\t", i + 1,
p[i]);

```

```

        if(allocation[i]      !=      -1)
printf("%d\n", allocation[i] + 1); else
printf("Not Allocated\n");
    }

```

```

// WORST FIT

```

```

for(i = 0; i < n; i++)
temp[i] = b[i];

```

```

for(i = 0; i < m; i++)
allocation[i] = -1;   for(i = 0; i <
m; i++)
{
    int worst
= -1;

```

```

        for(j = 0; j < n; j++)
        {
            if(temp[j]
>= p[i])
            {
                if(worst == -1 || temp[j] > temp[worst])
                {
worst = j;
                }
            }
        }

```

```

        if(worst != -1)
        {
            allocation[i] =
worst;
            temp[worst] =
-1;
        }
    }

```

```

    printf("\n--- Worst Fit ---\n");
    printf("Process
No.\tProcess Size\tBlock No.\n");

```

```

    for(i = 0; i < m; i++)
    {
        printf("%d\t\t%d\t\t", i + 1,
p[i]);
        if(allocation[i] != -1)
printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
    }

```

```

    return 0;
}

```

OUTPUT:

```
"C:\Users\Admin\Desktop\1W X + v

NAME : R NANDINI      USN : 1BM25CS497      SECTION : 'O'

Enter number of memory blocks: 5
Enter sizes of 5 memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of 4 processes:
212 417 112 426

--- First Fit ---
Process No.    Process Size    Block No.
1              212          2
2              417          5
3              112          3
4              426        Not Allocated

--- Best Fit ---
Process No.    Process Size    Block No.
1              212          4
2              417          2
3              112          3
4              426          5

--- Worst Fit ---
Process No.    Process Size    Block No.
1              212          5
2              417          2
3              112          4
4              426        Not Allocated

Process returned 0 (0x0)  execution time : 51.544 s
Press any key to continue.
```

LAB PROGRAM – 7

Question:

Write a C program to simulate page replacement algorithms. a)

FIFO

b) LRU

c) Optimal

CODE:

```
#include <stdio.h>

int main()
{   printf("\n\tNAME : R NANDINI\tUSN : 1BM25CS497\tSECTION :
'O'\n\n");
    int pages[50], frames[10];
    int n, f;
    int i, j, k;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // ----- FIFO -----
```

```

    int faults = 0;
    int next = 0;    int
    found;

    for(i = 0; i < f; i++)
    {
        frames[i]
    = -1;
    }

    printf("\n--- FIFO Page Replacement ---\n");

    for(i = 0; i < n; i++)
    {
        found
    = 0;

        for(j = 0; j < f; j++)
        {
            if(frames[j] ==
pages[i])
            {
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            frames[next] = pages[i];
            next = (next + 1) % f;
            faults++;

```

```

    }

    printf("Page %d -> [", pages[i]);

    for(j = 0; j < f; j++)
    {
        if(frames[j] != -1)
        {
            printf("%d",
frames[j]);

            if(j != f - 1)
            {
                printf("
");
            }
        }
    }

    printf("]\n");
}

printf("Total Page Faults (FIFO): %d\n", faults);

// ----- LRU -----

int time[10];
int count = 0;
faults = 0;

for(i = 0; i < f; i++)
{

```

```

        frames[i] = -1;
time[i] = 0;
    }

    printf("\n--- LRU Page Replacement ---\n");

    for(i = 0; i < n; i++)
    {
        found
= 0;

        for(j = 0; j < f; j++)
        {
            if(frames[j] ==
pages[i])
            {
                found
= 1;
                count++;
time[j] = count;
break;
            }
        }

        if(found == 0)
        {
            int
pos = -1;

            for(j = 0; j < f; j++)
            {
if(frames[j] == -1)
            {
pos = j;
break;

```

```

        }
    }

    if(pos == -1)
    {
pos = 0;
        for(j = 1; j < f;
j++)
        {
            if(time[j]
< time[pos])
            {
pos = j;
            }
        }
    }

    frames[pos] = pages[i];
count++;    time[pos] =
count;    faults++;
    }

    printf("Page %d -> [", pages[i]);

    for(j = 0; j < f; j++)
    {
if(frames[j] != -1)
        {
            printf("%d",
frames[j]);    if(j != f - 1)
        {
            printf(" ");
        }
    }
}

```

```

    }
}

printf("]\n");
}

printf("Total Page Faults (LRU): %d\n", faults);

// ----- OPTIMAL -----

faults = 0;

for(i = 0; i < f; i++)
{
    frames[i]
= -1;
}

printf("\n--- Optimal Page Replacement ---\n");

for(i = 0; i < n; i++)
{
    found
= 0;

    for(j = 0; j < f; j++)
    {
        if(frames[j] == pages[i])
        {
            found = 1;
            break;
        }
    }
}

```

```

    }

    if(found == 0)
    {
        int
pos = -1;

        for(j = 0; j < f; j++)
        {
            if(frames[j] == -1)
            {
                pos = j;
                break;
            }
        }

        if(pos == -1)
        {
            int
farthest = -1;          int
index;

            for(j = 0; j < f;
j++)
            {
                int
found_future = 0;

                for(k = i + 1; k < n; k++)
                {
                    if(frames[j] == pages[k])
                    {
                        if(k > farthest)
                        {
                            farthest =
k;
                            pos = j;

```

```

        }
        found_future =
1;        break;
    }
}

    if(found_future == 0)
    {
pos = j;
break;
    }
}
}

    frames[pos] = pages[i];
faults++;
}

    printf("Page %d -> [", pages[i]);

    for(j = 0; j < f; j++)
    {
        if(frames[j] != -1)
        {
            printf("%d",
frames[j]);
            if(j != f - 1)
            {
                printf("
");
            }
        }
    }
}

```

```

    }

    printf("]\n");
}

printf("Total Page Faults (Optimal): %d\n", faults);

return 0;
}

```

OUTPUT:

```

C:\Users\Admin\Desktop\IW X + v
NAME : R NANDINI USN : 1BM25CS497 SECTION : 'O'
Enter number of pages: 12
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 7 -> [7 0 -]
Page 0 -> [7 0 -]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]
Total Page Faults (FIFO): 10

--- LRU Page Replacement ---
Page 7 -> [7 0 -]
Page 0 -> [7 0 -]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]
Total Page Faults (LRU): 9

--- Optimal Page Replacement ---
Page 7 -> [7 0 -]
Page 0 -> [7 0 -]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 4 3]
Page 4 -> [2 4 3]
Page 2 -> [2 4 3]
Page 3 -> [0 4 3]
Page 0 -> [0 4 3]
Page 3 -> [0 4 3]
Total Page Faults (Optimal): 7

Process returned 0 (0x0) execution time : 57.823 s
Press any key to continue.

```

